

INFORMATION TECHNOLOGY INSTITUTE

LAB 2

Gain Linearization





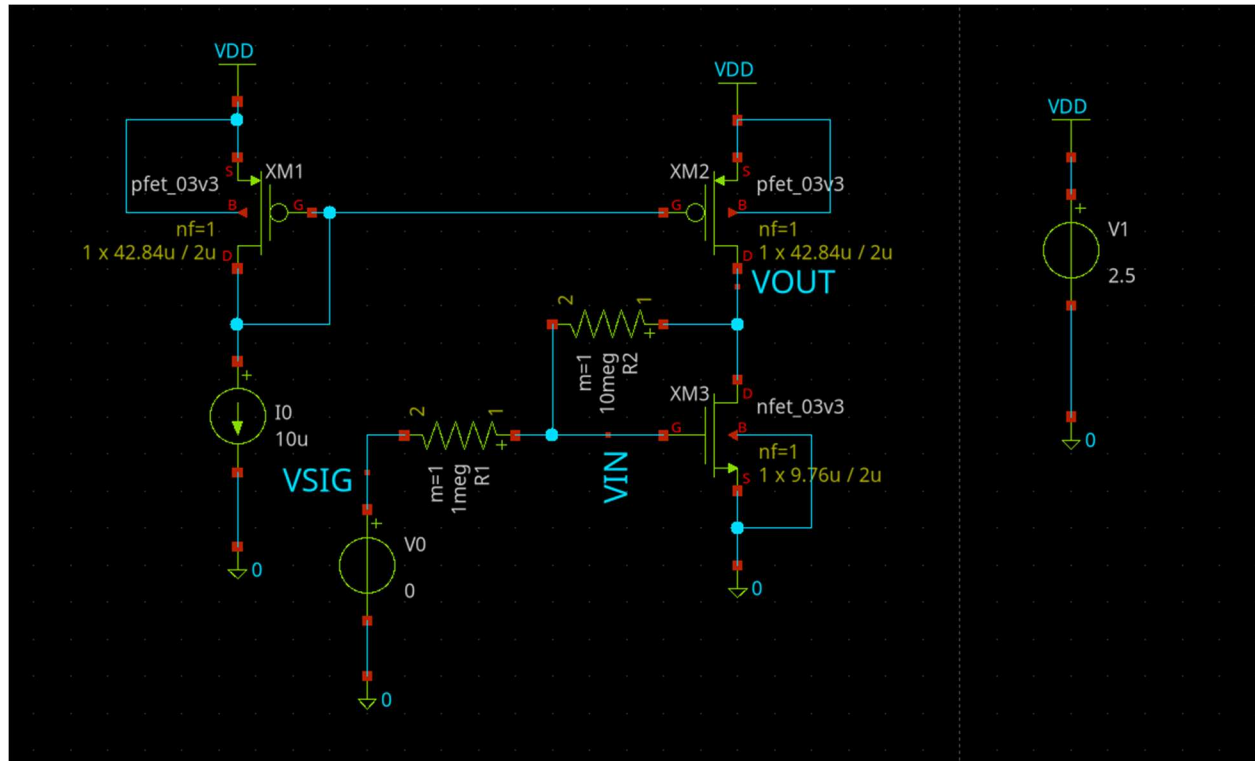
Contents

Negative Feedback	3
schematic	3
Set $R_F = 10\text{meg}$ and $R_{in} = 1\text{meg}$	3
Perform a DC sweep for the input voltage	4
Is V_{OUT} vs V_{SIG} linear in the operating range of the amplifier? Why?	4
The derivative of V_{OUT} vs V_{SIG}	5
V_{in} being constant	5
DC input range in which the gain is linear	6
Analytical :	6
Simulation:	6
transient simulation and report gm vs time.	7
Comment	8



Negative Feedback

schematic



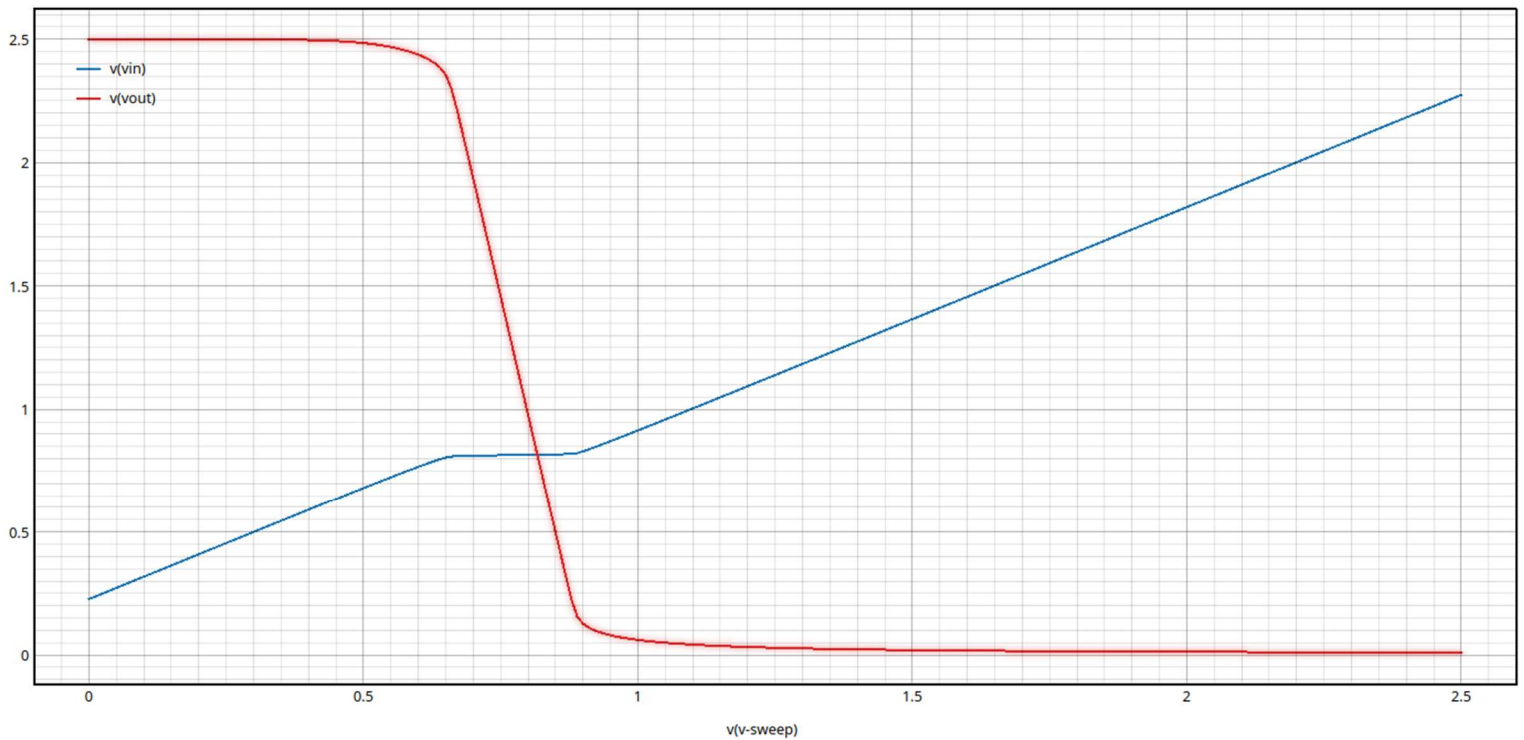
Set $R_f = 10\text{meg}$ and $R_{in} = 1\text{meg}$

As the feedback resistor (R_f) should be larger than R_{out} to avoid reducing the amplifier open-loop gain, so we choose 10meg . Choose R_{in} to give a voltage gain approximately equal to $R_f/R_{in} = |A_v|$ as given in the specs.



Perform a DC sweep for the input voltage

VIN and VOUT vs VSIG



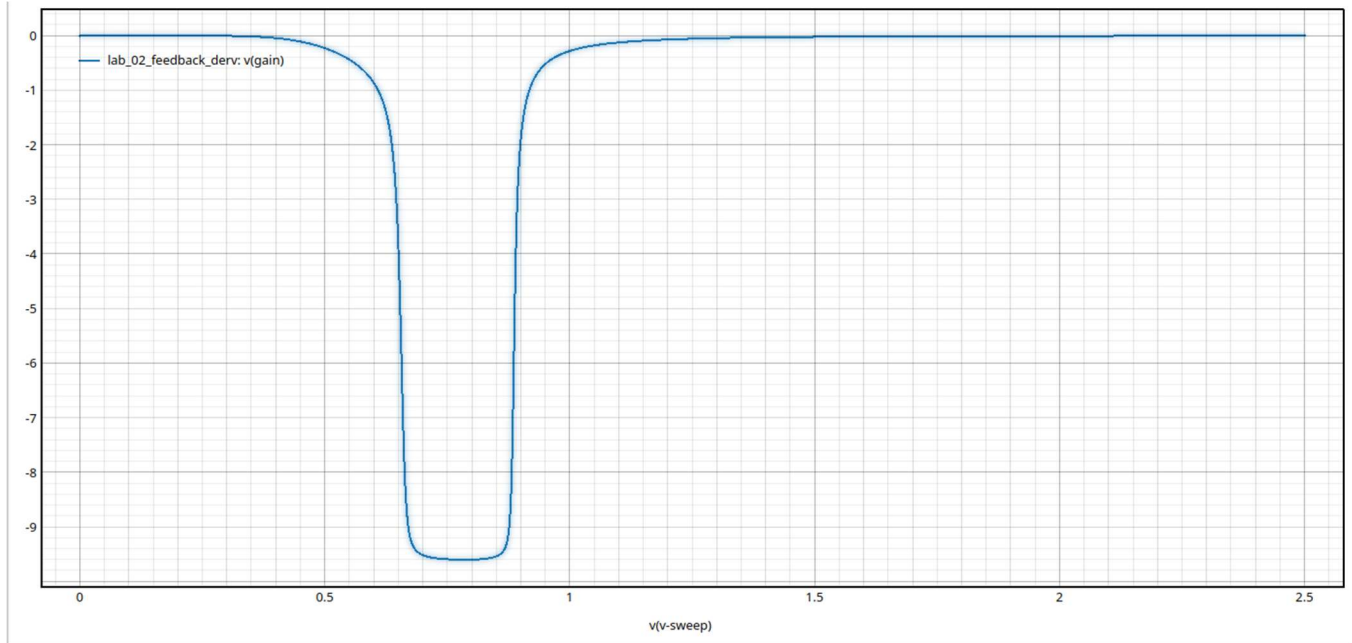
Is VOUT vs VSIG linear in the operating range of the amplifier? Why?

Yes, (V_{OUT}) versus (V_{SIG}) is approximately linear within the operating range of about (0.67 V) to (0.87 V). The slope remains nearly constant, changing only from approximately -9.1 to -9.38 V/V. This means the amplifier provides an almost constant inverting gain.

It is not perfectly linear because the MOSFET parameters, such as (g_m) and (r_o), change slightly with the input voltage. Outside this range, the transistor approaches cutoff or leaves saturation, causing the gain to drop and the response to become nonlinear.



The derivative of VOUT vs VSIG



The derivative $\frac{dV_{OUT}}{dV_{SIG}}$, which represents the small-signal gain, is approximately -9.1 to -9.38 V/V within the operating range.

The gain is approximately linear because the derivative is almost flat over this range, meaning it is nearly independent of V_{SIG} . The negative sign shows that the amplifier is inverting.

The gain is not perfectly constant because the MOSFET parameters change slightly with its operating point. Outside the operating range, the transistor approaches cutoff or leaves saturation, so the gain rapidly decreases toward zero and the output becomes nonlinear.

Vin being constant

is almost constant at approximately **0.815 V**, varying only from about **0.811 V** to **0.819 V**. This happens because the negative feedback forces V_{IN} to remain close to the transistor's required bias voltage. When V_{SIG} changes, V_{OUT} changes in the opposite direction through the feedback resistor, keeping V_{IN} nearly fixed. The small variation is caused by the amplifier's finite open-loop gain.



DC input range in which the gain is linear

Analytical :

For both transistors to remain in saturation:

$$V^* \leq V_{OUT} \leq V_{DD} - V^*$$

Using $V_{DD} = 2.5$ V and $V^* \approx 0.25$ V:

$$0.25 \leq V_{OUT} \leq 2.25 \text{ V}$$

Therefore, the analytical input-range width is

$$\Delta V_{SIG} \approx \frac{V_{DD} - 2V^*}{|A_v|} = \frac{2.5 - 2(0.25)}{10} = 0.20 \text{ V}$$

Since $V_{IN} \approx V_{GS1} \approx 0.815$ V, applying KCL gives

$$V_{OUT} = 11V_{GS} - 10V_{SIG}$$

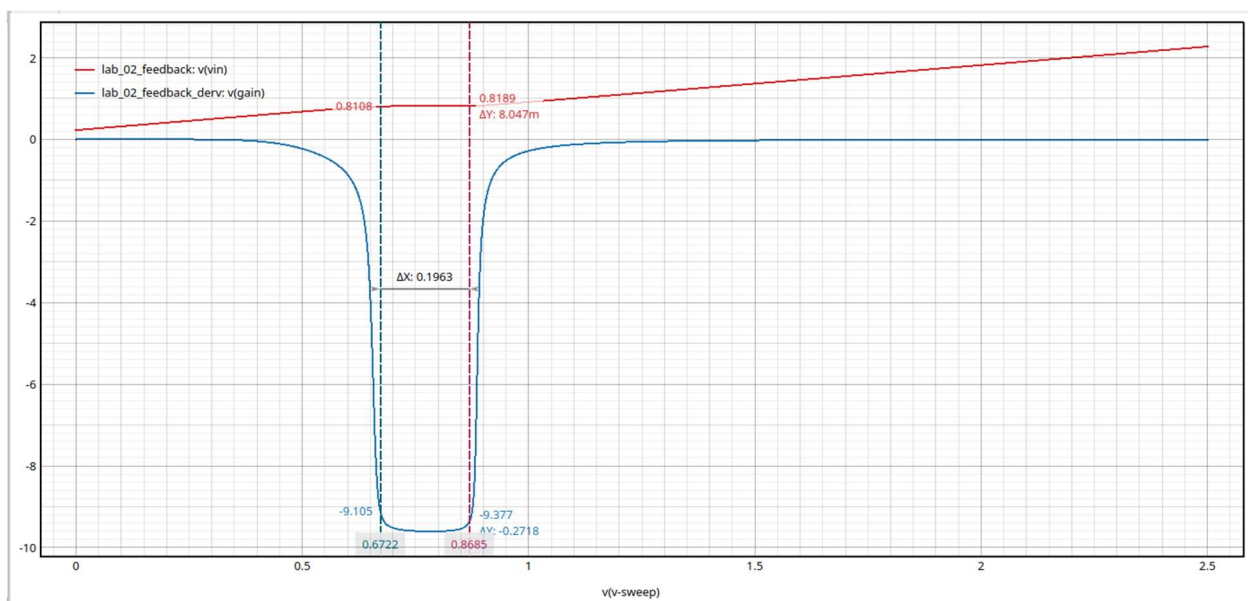
Thus,

$$V_{SIG,min} = \frac{11(0.815) - 2.25}{10} \approx 0.672 \text{ V}$$

$$V_{SIG,max} = \frac{11(0.815) - 0.25}{10} \approx 0.872 \text{ V}$$

$$0.672 \text{ V} \lesssim V_{SIG} \lesssim 0.872 \text{ V}$$

Simulation:



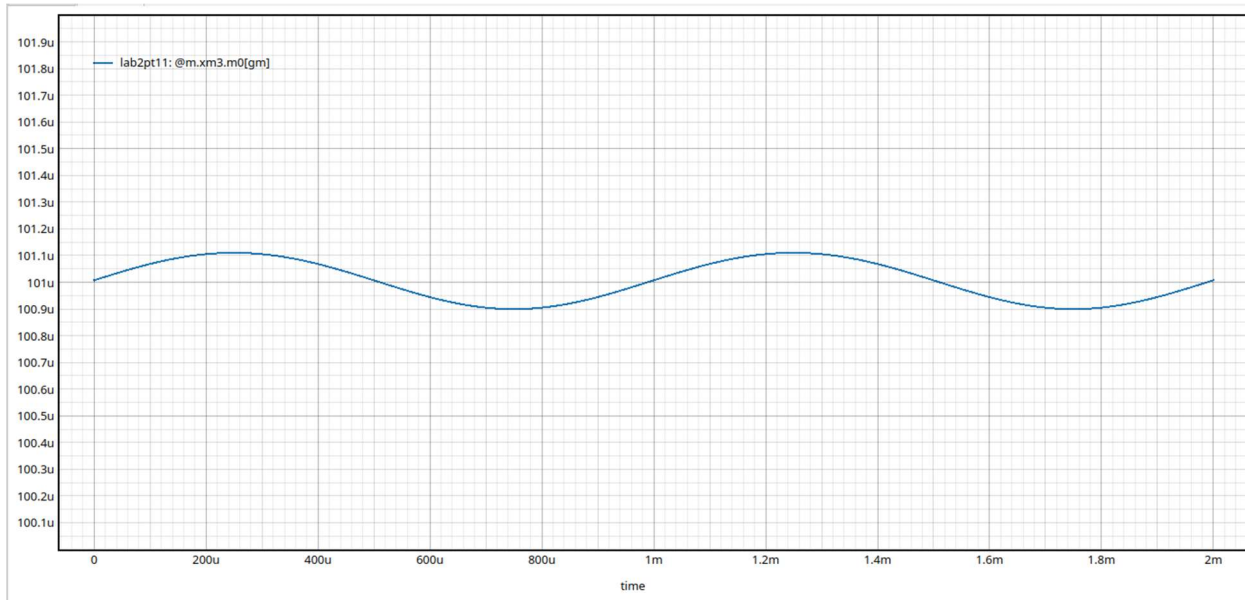


$$0.6722 \text{ V} \lesssim V_{SIG} \lesssim 0.8685 \text{ V}$$

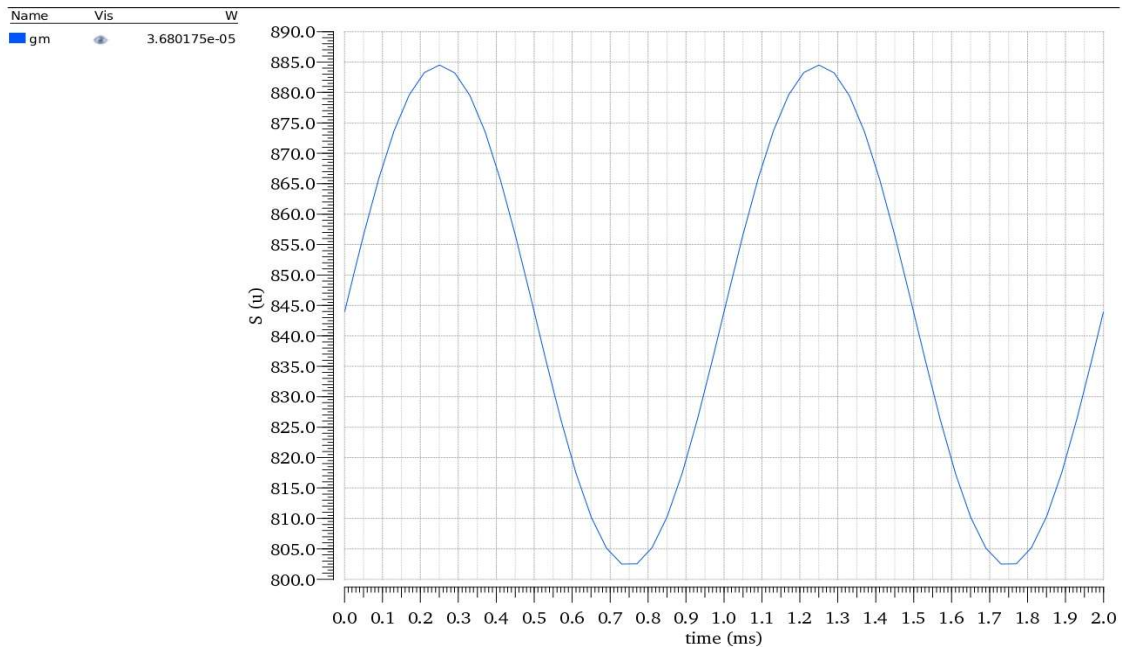
This due to the gain used in analytical =10 while the real gain is around 9.

transient simulation and report gm vs time.

With feedback



Without feedback





Comment

With negative feedback, g_m remains almost constant, varying only from approximately $100.8\ \mu\text{S}$ to $101.1\ \mu\text{S}$, or about 0.3% peak-to-peak. In the resistive-loaded open-loop amplifier, g_m changes much more, from approximately $802\ \mu\text{S}$ to $884\ \mu\text{S}$.

This happens because negative feedback keeps the transistor's V_{GS} and drain current nearly constant, so g_m changes very little as the input sine wave varies. Therefore, the feedback amplifier is more linear and produces less distortion than the open-loop amplifier.